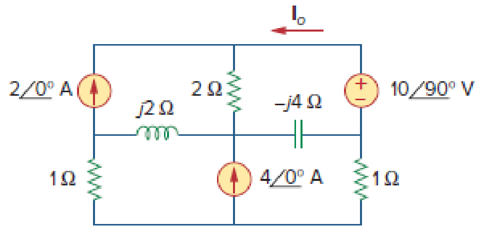
Chapter 10, Problem 38P

Problem

Using mesh analysis, obtain **I**o in the circuit shown in Fig. 10.83.

Figure 10.83:



Step-by-step solution

Step 1 of 6

Refer to figure 10.83 given in the textbook.

Redraw the circuit with mesh currents as shown in figure 1.

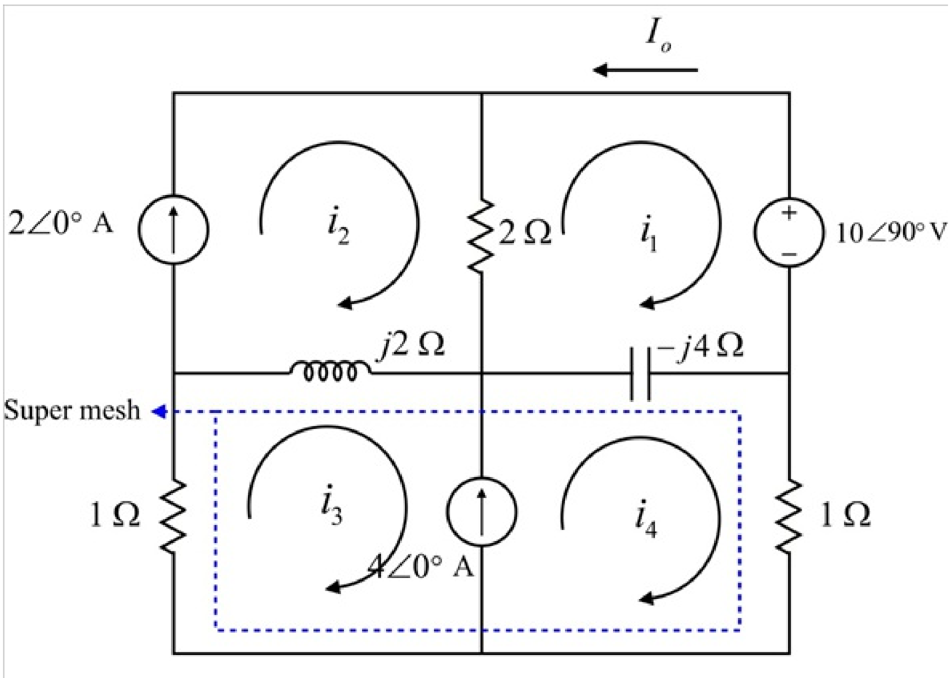


Figure 1

Step 2 of 6

Apply Kirchhoff's voltage law to mesh-1 with mesh current i_1 shown in figure 1.

$$10\angle 90^\circ + (-j4)(i_1 - i_4) + 2(i_1 - i_2) = 0$$

$$10j + i_1(-j4+2) + i_4 4j - 2i_2 = 0$$

Substitute $2\angle 0^\circ$ A for i_2.

$$10j + i_1(-j4+2) + i_4 4j - 2(2) = 0$$

$$i_1(2-j4) + i_4(j4) = 4 - j10 \quad \ldots\ldots (1)$$

Step 3 of 6

Since, there is a current source in common to mesh-3 and mesh-4 super mesh technique is used.

Write the super-mesh equation.

$$i_3 + j2(i_3 - i_2) - j4(i_4 - i_1) + i_4 = 0$$
$$i_1(4j) - i_2(2j) + i_3(1+2j) + i_4(1-4j) = 0$$

Substitute $2\angle 0^{\circ}\ \mathrm{A}$ for i_2.

$$i_1(4j) + i_3(1+2j) + i_4(1-4j) = 4j$$ …… (2)

Step 4 of 6

The super-mesh equation is,

$$i_4 - i_3 = 4\angle 0^{\circ}\ \mathrm{A}$$

$$i_3 = i_4 - 4$$

Substitute the expression of i_3 in equation (2).

$$i_1(j4) + (i_4 - 4)(1 + j2) + i_4(1 - j4) = j4$$
$$i_1(j4) + i_4(1 + j2) - 4 - j8 + i_4(1 - j4) = j4$$
$$i_1(j4) + i_4(1 + j2 + 1 - j4) = j12 + 4$$

$$i_1(j4) + i_4(2 - j2) = 4 + j12$$ …… (3)

Step 5 of 6

Recall equation (1).

$$i_1(2 - j4) + i_4(j4) = 4 - j10$$

Recall equation (3).

$$i_1(j4) + i_4(2 - j2) = 4 + 12j$$

The matrix form of equations (2) and (4) is,

$$\begin{bmatrix} 2-j4 & j4 \\ j4 & 2-j2 \end{bmatrix}\begin{bmatrix} i_1 \\ i_4 \end{bmatrix} = \begin{bmatrix} 4-j10 \\ 4+j12 \end{bmatrix}$$

Step 6 of 6

Calculate the following determinants.

$$\Delta = \begin{vmatrix} 2-j4 & j4 \\ j4 & 2-j2 \end{vmatrix}$$
$$= (2-j4)(2-j2)-(j4)(j4)$$
$$= 4-j4-j8-8+16$$
$$= 12-j12$$

Calculate the value of Δ_1.

$$\Delta_1 = \begin{vmatrix} 4-j10 & j4 \\ 4+j12 & 2-j2 \end{vmatrix}$$
$$= (4-j10)(2-j2)-(j4)(4+j12)$$
$$= 8-j8-j20-20-j16+48$$
$$= 36-44j$$

Calculate the current i_1.

$$i_1 = \frac{\Delta_1}{\Delta}$$

Substitute $36-44j$ for Δ_1, and $12-j12$ for Δ.

$$i_1 = \frac{36-44j}{12-j12}$$
$$= 3.33-j0.333 \text{ A}$$

Calculate current I_o.

$$I_o = -i_1$$

Substitute $4.3333+j0.6666 \text{ A}$ for i_1.

$$I_o = -(3.33-j0.333 \text{ A})$$
$$= -3.33+j0.333 \text{ A}$$
$$= 3.349\angle 174.28^\circ \text{ A}$$

Therefore, the value of current I_o is $\boxed{3.349\angle 174.28^\circ \text{ A}}$.